

Roberto André Mossi Milla

Software Engineer | AI Systems | Full-Stack | Computer Vision

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Summary

Software Engineer with experience building production AI systems, full-stack applications, computer-vision workflows, and backend services. Delivers end-to-end systems combining multimodal AI, OCR, cloud infrastructure, databases, and user-facing applications across healthcare, insurance, and industrial automation. Published AI researcher with hands-on experience in evolutionary computation, GPU-accelerated simulation, and production software engineering.

Experience

Ingeniería Digital

May 2026 – Present

Software Engineer (AI Systems)

Honduras

- Built an AI-assisted medical documentation platform from the ground up, reducing report preparation from approximately one week to 1–2 days and generating finalized documents in under one minute.
- Architected secure full-stack services for document upload, retrieval, versioning, review, and editing using React, Next.js, TypeScript, PostgreSQL, Auth0, and AWS S3.
- Built OCR and language-processing pipelines using Amazon Textract and OpenAI APIs to transform unstructured clinical records into structured, reviewable documentation.
- Developed an AI vehicle-damage assessment workflow that reduced assessment time from more than one week to approximately 10 minutes while automatically generating damage analyses and insurance estimates.
- Integrated AWS Rekognition for vehicle localization, Gemini Flash for damage detection, VIN-history retrieval, and image-analysis services into an end-to-end insurance workflow.
- Built interactive Konva-based annotation and validation tools allowing AI detections to be corrected and persisted for evaluating different models, processing stages, and AI combinations.
- Developed Playwright-based data-acquisition services and maintained CI/CD and engineering standards through GitHub Actions, Husky, ESLint, Prettier, Commitizen, and automated quality checks.

Gannon University

Aug. 2024 – May 2025

Computer Vision Engineer — Industry-Sponsored Project

Erie, PA

- Built components of a real-time industrial traceability system using LiDAR-triggered image capture, Ultralytics YOLOv8 object detection, Dockerized services, TimescaleDB, and monitoring dashboards.
- Integrated computer-vision events with persistent storage and operational interfaces, enabling near-real-time tracking of packages and pallets entering an industrial facility.
- Developed dataset correction and model-retraining workflows, including interactive bounding-box editing and image-processing pipelines for improving training data quality.

Gannon University

Aug. 2024 – May 2025

Research Assistant — Evolutionary Artificial Intelligence

Erie, PA

- Conducted faculty-supervised research on autonomous multi-agent learning and evolutionary computation using NeuroEvolution of Augmenting Topologies (NEAT), investigating emergent behavior in dynamic three-dimensional environments.
- Independently implemented NEAT and the underlying genome representation in C++, developing procedural environments and evolutionary simulation workflows in Unreal Engine.
- Used CUDA and PyTorch to support GPU-accelerated simulation and experimentation, contributing to a peer-reviewed ASEE publication and a Sigma Xi research presentation at Penn State University.

Dinant

Jun. 2023 – Jul. 2023

Full-Stack Developer

Honduras

- Developed and deployed database-backed business applications using Oracle SQL, JavaScript, HTML, and CSS, automating internal workflows that previously required weeks of manual processing.
- Contributed to data modeling, reporting, filtering, digital signatures, PDF generation, and application workflows across production systems used by business teams.

Selected Project

NOIR

Jan. 2026 – Present

Founder & Software Engineer (In Development)

Remote

- Designing and developing a cross-platform movie discovery and continuity platform using native SwiftUI applications and a Rust backend built with Axum, PostgreSQL, Redis, authentication, and cloud services.
- Architecting services for movie-metadata aggregation, availability tracking, synchronization, and cross-device access, with focused AI assistance for natural-language movie discovery.
- Owning product and engineering decisions across UX, data modeling, backend implementation, testing workflows, internationalization, and deployment planning.

Open Source

Copilot-Lualine

2024 – Present

Creator & Maintainer

Open Source

- Created and maintain an open-source Neovim plugin integrating GitHub Copilot status and controls into the lualine status bar; project has received 117+ GitHub stars and community contributions.

Technical Skills

Languages: Python, C++, Rust, TypeScript, JavaScript, SQL, Java, Bash, Lua

Artificial Intelligence: Multimodal AI, OpenAI APIs, Gemini Flash, AWS Rekognition, Amazon Textract, Computer Vision, OCR, Ultralytics YOLOv8, PyTorch, CUDA, NEAT, evolutionary computation

Application Engineering: React, Next.js, Node.js, SwiftUI, Axum, REST APIs, PostgreSQL, Redis, Oracle Database, Auth0, Playwright, Konva

Infrastructure & Delivery: Docker, AWS S3, GitHub Actions, Git, CI/CD, Linux, CMake, Husky, ESLint, Prettier, Commitizen

Data & Systems: TimescaleDB, event-driven services, database design, cloud storage, document-processing workflows

Education & Recognition

Gannon University

2025

B.S. Computer Systems Engineering

Erie, PA

- Peer-reviewed publication with the American Society for Engineering Education (ASEE); research presentation at Sigma Xi.